BlossomGifts - Online Flower and Gift Ordering System

INTERNET PROGRAMMING LAB

AIM

To design and develop an online flower and gift ordering website using HTML, CSS and JavaScript with a logically categorised catalogue (popular categories first), search and price filters, a shopping cart with customised delivery date and time, order tracking, and an admin module to add and remove items, modify prices and descriptions, and view the customers who checked out.

ALGORITHM

Step 1: Start. Load the catalogue (items, categories, sub-categories, stock) and the order list from localStorage; seed sample data on first run.

Step 2: Login: a customer enters a name; an admin selects the Admin role and gives the password. Show the Admin Panel tab only for the admin. Logout clears the session and the cart.

Step 3: Home page: count the units sold per category, sort the categories by popularity and display every category together with its sub-categories.

Step 4: Search and filter: filter the products by selected category / sub-category, keyword in the item name and the minimum and maximum price, and display each result with its price and availability.

Step 5: Display Item Details in a pop-up (description, category, price, stock). Add to Cart is allowed only for logged-in users and for items in stock.

Step 6: Cart: show the items with quantity edit (+ / -) and Remove. Compute the sub-total and apply a 5% loyalty discount when the customer has two or more earlier orders.

Step 7: Checkout: read recipient name, address, delivery date, delivery time slot, payment method and gift message; validate, create the order with status Placed, reduce the stock, increase the sold count, clear the cart and show the e-mail / SMS confirmation.

Step 8: My Orders: list the orders of the logged-in customer with a four-stage tracker (Placed, Packed, Out for Delivery, Delivered).

Step 9: Admin: add a new item, modify the price, stock and description of an item, remove an item, view every customer order with delivery address and move the order status to the next stage.

Step 10: Save every change to localStorage and re-render the page. Stop.

PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>BlossomGifts - Flowers &amp;amp; Gifts</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <div class="brand">&#127802; BlossomGifts</div>
  <div id="authBox">
    <input id="uName" placeholder="Your name">
    <select id="uRole"><option value="customer">Customer</option><option value="admin">Admin</option></select>
    <input id="uPass" type="password" placeholder="Admin password" class="hidden">
    <button id="loginBtn">Login</button>
  </div>
</header>
<nav id="menu">
  <button data-v="shop" class="active">Shop</button>
  <button data-v="cart">Cart (<span id="cartCount">0</span>)</button>
  <button data-v="orders">My Orders</button>
  <button data-v="admin" id="adminTab" class="hidden">Admin Panel</button>
</nav>
<main>
  <section id="shop" class="view active">
    <div id="cats"></div>
    <div class="filters">
      <input id="q" placeholder="Search flowers or gifts...">
      <label>Price Rs. <input id="pmin" type="number" placeholder="min"> - <input id="pmax" type="number"
          placeholder="max"></label>
      <button id="clearF">Clear</button>
    </div>
    <div id="products" class="cards"></div>
  </section>

  <section id="cart" class="view">
    <h2>Shopping Cart</h2>
    <div id="cartBody"></div>
    <div id="checkout" class="box hidden">
      <h3>Delivery Details</h3>
      <div class="grid">
        <label>Recipient Name<input id="rName"></label>
        <label>Delivery Address<input id="rAddr"></label>
        <label>Delivery Date<input type="date" id="rDate"></label>
        <label>Delivery Time<select id="rTime"><option>9 AM - 12 PM</option><option>12 PM - 3
            PM</option><option>3 PM - 6 PM</option><option>6 PM - 9 PM</option></select></label>
        <label>Payment<select id="rPay"><option>UPI</option><option>Credit / Debit Card</option><option>Net
            Banking</option><option>Cash on Delivery</option></select></label>
        <label>Gift Message<input id="rMsg" placeholder="Message on the card"></label>
      </div>
      <button id="placeOrder" class="primary">Place Order</button>
    </div>
    <p id="cartMsg" class="msg"></p>
  </section>

  <section id="orders" class="view"><h2>My Orders &amp;amp; Tracking</h2><div id="ordBody"></div></section>

  <section id="admin" class="view">
    <h2>Admin Panel</h2>
    <div class="box"><h3>Add New Item</h3>
      <div class="grid">
        <label>Name<input id="aName"></label>
        <label>Category<select id="aCat"></select></label>
        <label>Sub-category<input id="aSub" placeholder="e.g. Roses"></label>
        <label>Price (Rs.)<input id="aPrice" type="number"></label>
        <label>Stock<input id="aStock" type="number" value="10"></label>
        <label>Description<input id="aDesc"></label>
      </div>
      <button id="addItem" class="primary">Add Item</button>
      <span id="admMsg" class="msg"></span>
    </div>
    <h3>Manage Inventory (modify price / description, remove)</h3>
    <div id="invBody"></div>
    <h3>Customer Orders (customers who checked out)</h3>
    <div id="admOrders"></div>
  </section>
</main>
<div id="modal" class="hidden"><div id="modalBox"></div></div>
<script src="script.js"></script>
```

```
</body>
</html>
```

style.css

```
:root{--pink:#c2185b;--dark:#7b1244;--soft:#fff0f6;--line:#f1c9da}
*{box-sizing:border-box}
body{margin:0;font-family:"Trebuchet MS",Arial,sans-serif;background:#fffafc;color:#33202a}
header{display:flex;justify-content:space-between;align-items:center;flex-wrap:wrap;gap:8px;background:linear-
    gradient(90deg,#7b1244,#e0527f);color:#fff;padding:12px 24px}
.brand{font-size:26px;font-weight:700}
#authBox{display:flex;gap:6px;align-items:center}
#authBox input,#authBox select{padding:6px;border-radius:5px;border:0}
#authBox button,nav button{cursor:pointer;border:0;border-radius:5px;padding:7px 14px;font:inherit}
#authBox button{background:#fff;color:var(--dark);font-weight:700}
nav{display:flex;gap:4px;background:var(--soft);padding:6px 24px;border-bottom:1px solid var(--line)}
nav button{background:transparent;color:var(--dark)}
nav button.active{background:var(--pink);color:#fff}
main{max-width:1050px;margin:16px auto;padding:0 14px}
.view{display:none}.view.active{display:block}
h2,h3{color:var(--dark)}
.cat{display:inline-block;margin:4px 6px 4px 0}
.cat>button{background:#fff;border:2px solid var(--pink);color:var(--dark);border-radius:22px;padding:7px
    16px;cursor:pointer;font-weight:700}
.cat>button.sel{background:var(--pink);color:#fff}
.subs{margin:2px 0 0 10px;font-size:12px}
.subs span{display:inline-block;background:#fde3ee;border-radius:10px;padding:2px 8px;margin:2px 2px 0
    0;cursor:pointer}
.subs span.sel{background:var(--dark);color:#fff}
.filters{display:flex;gap:10px;flex-wrap:wrap;margin:12px 0;align-items:center}
.filters input{padding:8px;border:1px solid #d9a6bd;border-radius:6px}
.filters label input{width:80px}
.cards{display:grid;grid-template-columns:repeat(auto-fill,minmax(210px,1fr));gap:14px}
.card{background:#fff;border:1px solid var(--line);border-radius:12px;overflow:hidden;box-shadow:0 2px 6px
    rgba(123,18,68,.08)}
.pic{height:110px;display:flex;align-items:center;justify-content:center;font-size:54px;background:linear-
    gradient(135deg,#ffe3ee,#ffd0e1)}
.card .in{padding:10px 12px}
.card h4{margin:0 0 4px}.tag{font-size:11px;color:#8a5a6e}
.price{color:var(--pink);font-weight:700;font-size:18px}
.stock{font-size:12px}.ok{color:#2a7a3b}.out{color:#c0392b}
button.primary,.btn{background:var(--pink);color:#fff;border:0;border-radius:6px;padding:7px
    12px;cursor:pointer;font:inherit}
.btn.alt{background:#fff;color:var(--dark);border:1px solid var(--pink)}
.btn.red{background:#c0392b}
.box{background:#fff;border:1px solid var(--line);border-radius:10px;padding:14px;margin:12px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(210px,1fr));gap:10px;margin-bottom:10px}
label{display:flex;flex-direction:column;font-size:13px;font-weight:700;gap:3px}
.grid input,.grid select,.box input,.box select{padding:7px;border:1px solid #d9a6bd;border-
    radius:6px;font:inherit}
table{width:100%;border-collapse:collapse;font-size:13px;background:#fff}
th,td{border-bottom:1px solid var(--line);padding:7px;text-align:left}
th{background:#fde3ee}
td input{width:80px;padding:4px}
.hidden{display:none!important}
.msg{font-weight:700;color:#2a7a3b}.msg.err{color:#c0392b}
.track{display:flex;gap:4px;margin:6px 0}
.track div{flex:1;text-align:center;font-size:11px;padding:5px;background:#eee;border-radius:4px}
.track .done{background:#2a7a3b;color:#fff}
#modal{position:fixed;inset:0;background:rgba(0,0,0,.45);display:flex;align-items:center;justify-content:center}
#modalBox{background:#fff;padding:20px;border-radius:12px;width:min(420px,92vw)}
```

script.js

```
const CATS = ['Birthday','Anniversary','Wedding','Get Well Soon','Congratulations'];
const STEPS = ['Placed','Packed','Out for Delivery','Delivered'];
const $ = id => document.getElementById(id);
const seed = () => ({
  items:[
   {id:1,name:'Red Rose
       Bouquet',cat:'Anniversary',sub:'Roses',price:899,stock:12,sold:40,icon:'\u{1F339}',desc:'24 fresh red
       roses wrapped in premium paper.'},
   {id:2,name:'Birthday Tulip
       Mix',cat:'Birthday',sub:'Bouquets',price:749,stock:8,sold:55,icon:'\u{1F337}',desc:'Colourful tulips with
        a birthday greeting card.'},
   {id:3,name:'Chocolate Gift Basket',cat:'Birthday',sub:'Gift
       Baskets',price:1299,stock:5,sold:30,icon:'\u{1F36B}',desc:'Assorted chocolates in a handcrafted
       basket.'},
   {id:4,name:'Teddy with Roses',cat:'Anniversary',sub:'Personalised
       Gifts',price:1499,stock:0,sold:25,icon:'\u{1F9F8}',desc:'Soft teddy bear with 12 pink roses.'},
   {id:5,name:'Bridal Orchid
       Bouquet',cat:'Wedding',sub:'Bouquets',price:2499,stock:4,sold:12,icon:'\u{1F490}',desc:'Elegant white
       orchid bridal bouquet.'},
   {id:6,name:'Sunflower Cheer Vase',cat:'Get Well
       Soon',sub:'Vases',price:599,stock:15,sold:18,icon:'\u{1F33B}',desc:'Bright sunflowers in a glass vase.'},
   {id:7,name:'Congratulations Hamper',cat:'Congratulations',sub:'Gift
       Baskets',price:1799,stock:6,sold:9,icon:'\u{1F381}',desc:'Fruits, dry fruits and flowers hamper.'}],
  orders:[
   {id:101,user:'Ravi',items:[{id:2,name:'Birthday Tulip Mix',price:749,qty:1}],total:749,addr:'Anna Nagar,
       Chennai',date:'2026-10-05',slot:'12 PM - 3 PM',pay:'UPI',status:3},
   {id:102,user:'Ravi',items:[{id:1,name:'Red Rose Bouquet',price:899,qty:2}],total:1798,addr:'Anna Nagar,
       Chennai',date:'2026-10-12',slot:'9 AM - 12 PM',pay:'Credit / Debit Card',status:3}],
  nextItem:8,nextOrder:103});
let db = JSON.parse(localStorage.getItem('blossom') || 'null') || seed();
const save = () => localStorage.setItem('blossom', JSON.stringify(db));
let user = null, cart = [], fCat = '', fSub = '';

/* ---------- login / logout ---------- */
$('uRole').onchange = () => $('uPass').classList.toggle('hidden', $('uRole').value !== 'admin');
$('loginBtn').onclick = () => {
  if(user){ user = null; cart = []; render(); return; }                      // logout
  const n = $('uName').value.trim(), role = $('uRole').value;
  if(!n) return alert('Please enter your name');
  if(role === 'admin' && $('uPass').value !== 'admin123') return alert('Wrong admin password (hint: admin123)');
  user = {name:n, role}; render();
};
document.querySelectorAll('#menu button').forEach(b => b.onclick = () => show(b.dataset.v));
function show(v){
  document.querySelectorAll('#menu button').forEach(b => b.classList.toggle('active', b.dataset.v === v));
  document.querySelectorAll('.view').forEach(s => s.classList.toggle('active', s.id === v));
  render();
}

/* ---------- shop: categories, filters, products ---------- */
function renderCats(){
  // most popular categories (by units sold) are listed first
  const pop = c => db.items.filter(i => i.cat === c).reduce((s,i) => s + i.sold, 0);
  const order = [...CATS].sort((a,b) => pop(b) - pop(a));
  $('cats').innerHTML = order.map(c => {
    const subs = [...new Set(db.items.filter(i => i.cat === c).map(i => i.sub))];
    return `<div class="cat"><button class="${fCat===c?'sel':''}" onclick="setCat('${c}','')">${c}</button>
      <div class="subs">${subs.map(s => `<span class="${fSub===s&&fCat===c?'sel':''}"
          onclick="setCat('${c}','${s}')">${s}</span>`).join('')}</div></div>`;
  }).join('');
}
function setCat(c, s){ fCat = (fCat === c && !s && !fSub) ? '' : c; fSub = s; renderShop(); }
['q','pmin','pmax'].forEach(id => $(id).oninput = renderShop);
$('clearF').onclick = () => { fCat = fSub = ''; $('q').value = $('pmin').value = $('pmax').value = '';
    renderShop(); };
function renderShop(){
  renderCats();
  const q = $('q').value.toLowerCase(), lo = +$('pmin').value || 0, hi = +$('pmax').value || 1e9;
  const list = db.items.filter(i => (!fCat || i.cat === fCat) && (!fSub || i.sub === fSub) &&
     i.name.toLowerCase().includes(q) && i.price >= lo && i.price <= hi);
  $('products').innerHTML = list.map(i => `<div class="card"><div class="pic">${i.icon}</div><div class="in">
    <h4>${i.name}</h4><div class="tag">${i.cat} &bull; ${i.sub}</div>
    <div class="price">Rs. ${i.price}</div>
    <div class="stock ${i.stock?'ok':'out'}">${i.stock ? 'In stock ('+i.stock+')' : 'Out of stock'}</div><br>
    <button class="btn alt" onclick="details(${i.id})">Details</button>
    <button class="btn" ${i.stock?'':'disabled'} onclick="addCart(${i.id})">Add to
        Cart</button></div></div>`).join('')
    || '<p>No items match your search.</p>';
}
function details(id){
  const i = db.items.find(x => x.id === id);
  $('modalBox').innerHTML = `<div class="pic">${i.icon}</div><h3>${i.name}</h3><p>${i.desc}</p>
```

```
    <p><b>Category:</b> ${i.cat} / ${i.sub}<br><b>Price:</b> Rs. ${i.price}<br><b>Availability:</b> ${i.stock ?
        i.stock + ' available' : 'Out of stock'}</p>
    <button class="btn" onclick="$('modal').classList.add('hidden')">Close</button>`;
  $('modal').classList.remove('hidden');
}

/* ---------- cart & checkout ---------- */
function addCart(id){
  if(!user) return alert('Please login to add items to the cart.');
  const it = db.items.find(i => i.id === id), line = cart.find(c => c.id === id);
  if(line) line.qty = Math.min(line.qty + 1, it.stock); else cart.push({id, name:it.name, price:it.price,
      qty:1});
  $('cartCount').textContent = cart.reduce((s,c) => s + c.qty, 0);
}
function changeQty(id, d){
  const line = cart.find(c => c.id === id), stock = db.items.find(i => i.id === id).stock;
  line.qty = Math.max(0, Math.min(stock, line.qty + d));
  cart = cart.filter(c => c.qty > 0); renderCart();
}
function loyalDiscount(){ return db.orders.filter(o => o.user === user.name).length >= 2 ? 0.05 : 0; }
function renderCart(){
  $('cartCount').textContent = cart.reduce((s,c) => s + c.qty, 0);
  if(!cart.length){ $('cartBody').innerHTML = '<p>Your cart is empty.</p>';
      $('checkout').classList.add('hidden'); return; }
  const sub = cart.reduce((s,c) => s + c.price*c.qty, 0), disc = user ? Math.round(sub*loyalDiscount()) : 0;
  $('cartBody').innerHTML = `<table><tr><th>Item</th><th>Price</th><th>Qty</th><th>Total</th><th></th></tr>` +
    cart.map(c => `<tr><td>${c.name}</td><td>Rs. ${c.price}</td>
      <td><button class="btn alt" onclick="changeQty(${c.id},-1)">-</button> ${c.qty} <button class="btn alt"
          onclick="changeQty(${c.id},1)">+</button></td>
      <td>Rs. ${c.price*c.qty}</td><td><button class="btn red"
          onclick="changeQty(${c.id},-99)">Remove</button></td></tr>`).join('') +
    `<tr><td colspan="3" align="right"><b>Loyalty discount</b></td><td>- Rs. ${disc}</td><td></td></tr>
     <tr><td colspan="3" align="right"><b>Grand Total</b></td><td><b>Rs. ${sub -
         disc}</b></td><td></td></tr></table>`;
  $('checkout').classList.remove('hidden');
}
$('placeOrder').onclick = () => {
  if(!$('rName').value || !$('rAddr').value || !$('rDate').value){ $('cartMsg').textContent = 'Please fill
      recipient, address and delivery date.'; $('cartMsg').className = 'msg err'; return; }
  const sub = cart.reduce((s,c) => s + c.price*c.qty, 0), total = Math.round(sub*(1 - loyalDiscount()));
  const o = {id:db.nextOrder++, user:user.name, items:cart.map(c => ({...c})), total, addr:$('rName').value + ',
       ' + $('rAddr').value,
            date:$('rDate').value, slot:$('rTime').value, pay:$('rPay').value, msg:$('rMsg').value, status:0};
  cart.forEach(c => { const it = db.items.find(i => i.id === c.id); it.stock -= c.qty; it.sold += c.qty; });
  db.orders.push(o); save(); cart = [];
  $('cartMsg').className = 'msg';
  $('cartMsg').textContent = `Order #${o.id} placed! Confirmation sent by email and SMS. Delivery on ${o.date}
      (${o.slot}).`;
  renderCart();
};

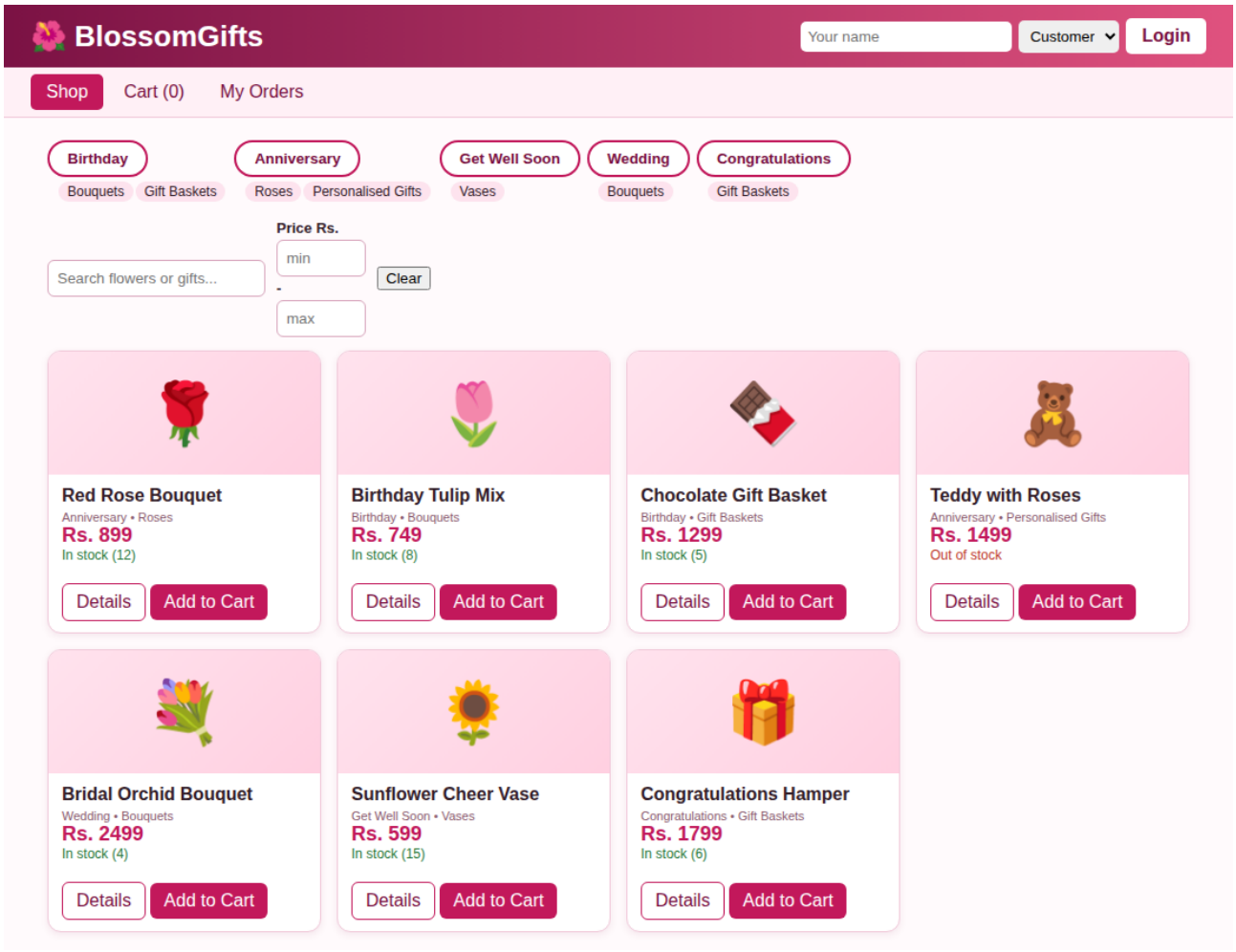
/* ---------- orders & tracking ---------- */
const track = s => `<div class="track">${STEPS.map((t,i) => `<div
    class="${i<=s?'done':''}">${t}</div>`).join('')}</div>`;
function renderOrders(){
  const mine = user ? db.orders.filter(o => o.user === user.name) : [];
  $('ordBody').innerHTML = mine.map(o => `<div class="box"><b>Order #${o.id}</b> - Rs. ${o.total} - ${o.pay}<br>
    ${o.items.map(i => i.name + ' x' + i.qty).join(', ')}<br>Delivery: ${o.date}, ${o.slot} to
        ${o.addr}${track(o.status)}</div>`).join('')
    || '<p>No orders yet. Login with the name "Ravi" to see sample order history.</p>';
}

/* ---------- admin ---------- */
CATS.forEach(c => $('aCat').add(new Option(c, c)));
$('addItem').onclick = () => {
  const n = $('aName').value.trim(), p = +$('aPrice').value;
  if(!n || p <= 0){ $('admMsg').textContent = 'Enter a name and valid price.'; $('admMsg').className = 'msg
      err'; return; }
  db.items.push({id:db.nextItem++, name:n, cat:$('aCat').value, sub:$('aSub').value || 'General', price:p,
      stock:+$('aStock').value,
                 sold:0, icon:'\u{1F381}', desc:$('aDesc').value});
  save(); $('admMsg').className = 'msg'; $('admMsg').textContent = 'Item added to catalogue.'; renderAdmin();
};
function updItem(id){
  const it = db.items.find(i => i.id === id);
  it.price = +$('p'+id).value; it.desc = $('d'+id).value; it.stock = +$('s'+id).value; save(); renderAdmin();
}
function delItem(id){ db.items = db.items.filter(i => i.id !== id); save(); renderAdmin(); }
function nextStatus(id){ const o = db.orders.find(x => x.id === id); if(o.status < 3) o.status++; save();
    renderAdmin(); }
function renderAdmin(){
  $('invBody').innerHTML = `<table><tr><th>ID</th><th>Item</th><th>Price
      (Rs.)</th><th>Stock</th><th>Description</th><th>Action</th></tr>` +
    db.items.map(i => `<tr><td>${i.id}</td><td>${i.icon} ${i.name}</td><td><input id="p${i.id}" type="number"
```

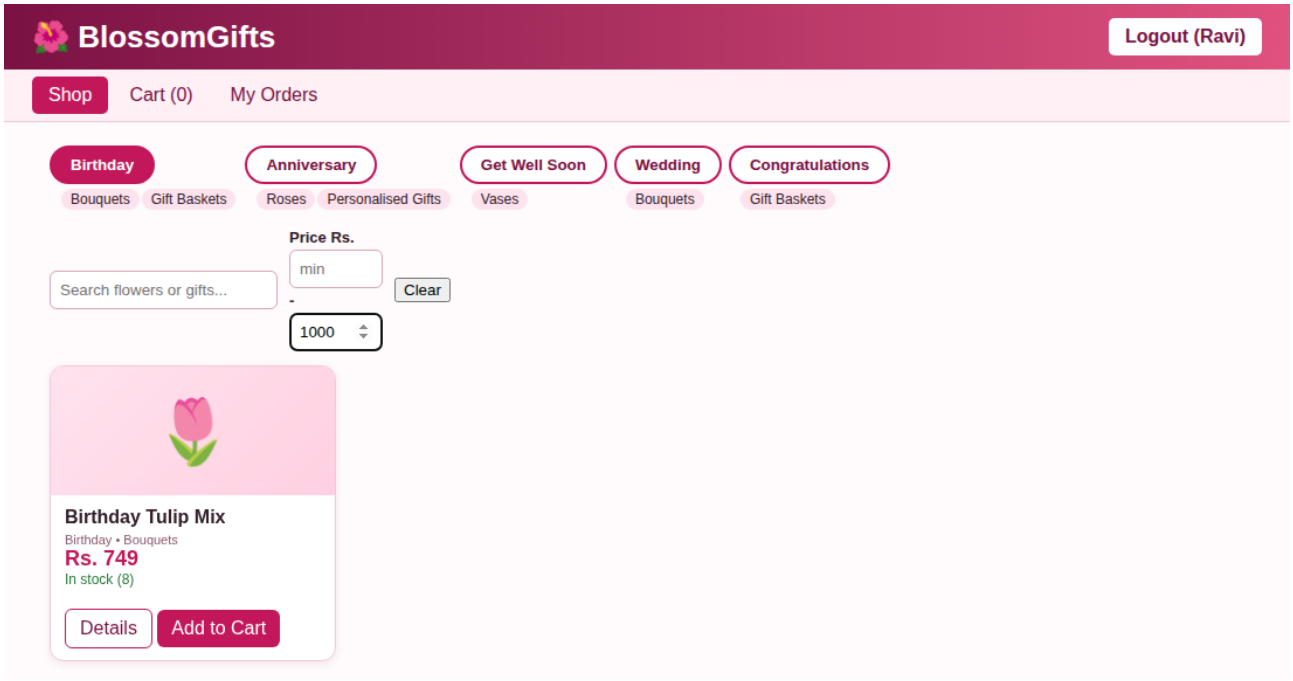
```
        value="${i.price}"></td>
      <td><input id="s${i.id}" type="number" value="${i.stock}"></td><td><input id="d${i.id}"
          style="width:170px" value="${i.desc}"></td>
      <td><button class="btn" onclick="updItem(${i.id})">Update</button> <button class="btn red"
          onclick="delItem(${i.id})">Remove</button></td></tr>`).join('') + '</table>';
  $('admOrders').innerHTML = `<table><tr><th>Order</th><th>Customer</th><th>Items</th><th>Total</th><th>Delivery
       address</th><th>Status</th></tr>` +
    db.orders.map(o => `<tr><td>#${o.id}</td><td>${o.user}</td><td>${o.items.map(i => i.name+' x'+i.qty).join(',
        ')}</td><td>Rs. ${o.total}</td>
      <td>${o.addr}</td><td>${STEPS[o.status]} <button class="btn alt"
          onclick="nextStatus(${o.id})">Next</button></td></tr>`).join('') + '</table>';
}

function render(){
  const logged = !!user;
  $('loginBtn').textContent = logged ? 'Logout (' + user.name + ')' : 'Login';
  ['uName','uRole','uPass'].forEach(id => $(id).classList.toggle('hidden', logged || (id === 'uPass' &&
      $('uRole').value !== 'admin')));
  $('adminTab').classList.toggle('hidden', !(logged && user.role === 'admin'));
  renderShop(); renderCart(); renderOrders(); if(logged && user.role === 'admin') renderAdmin();
}
render();
```

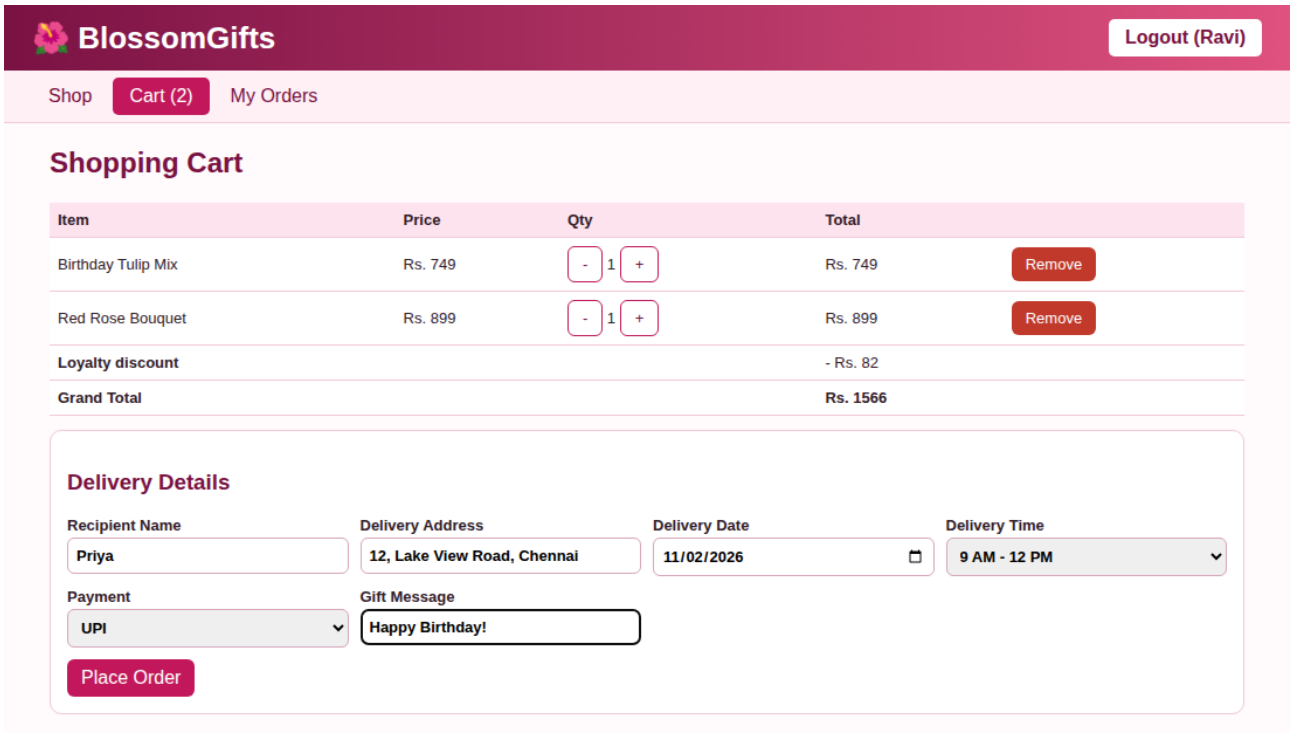
OUTPUT



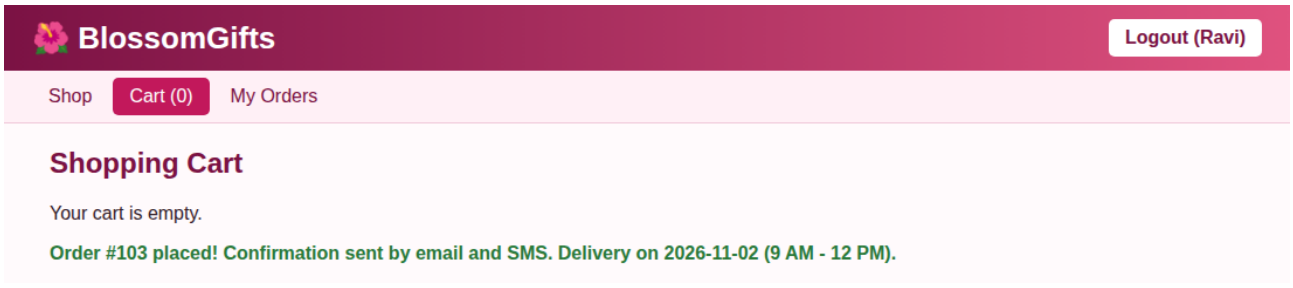
Home page: categories ordered by popularity with their sub-categories, search and price-range filters



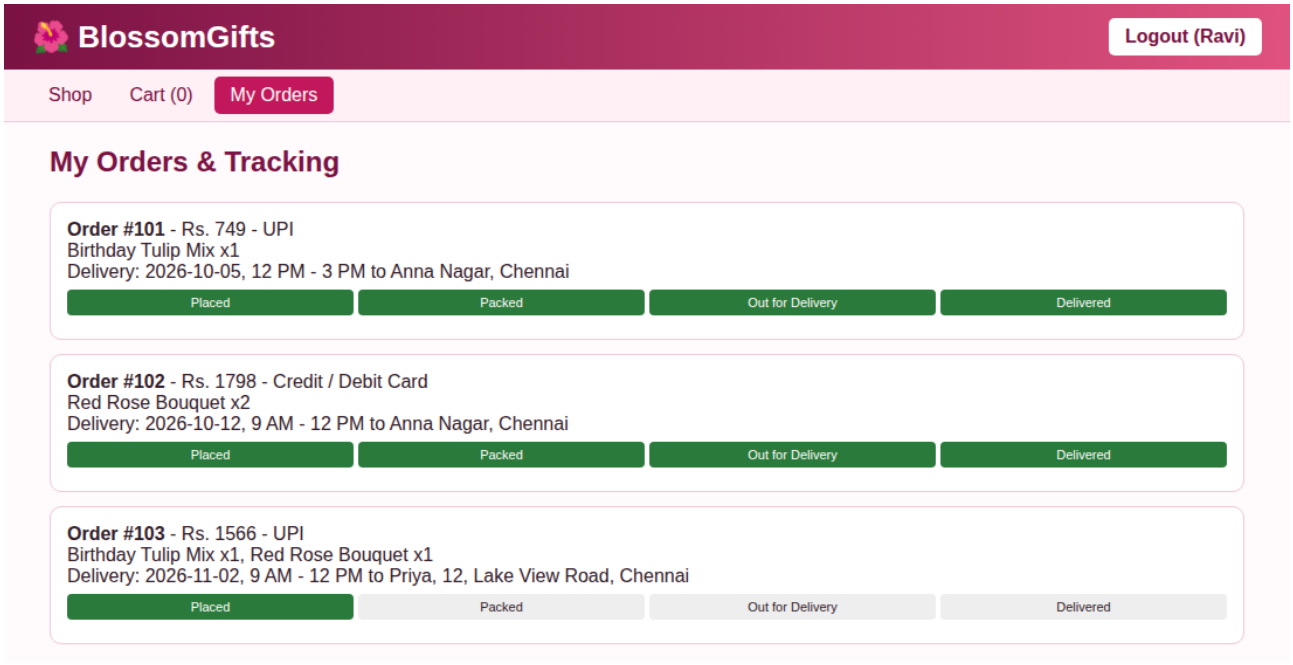
Customer login with Birthday category and price range (up to Rs. 1000) filter applied



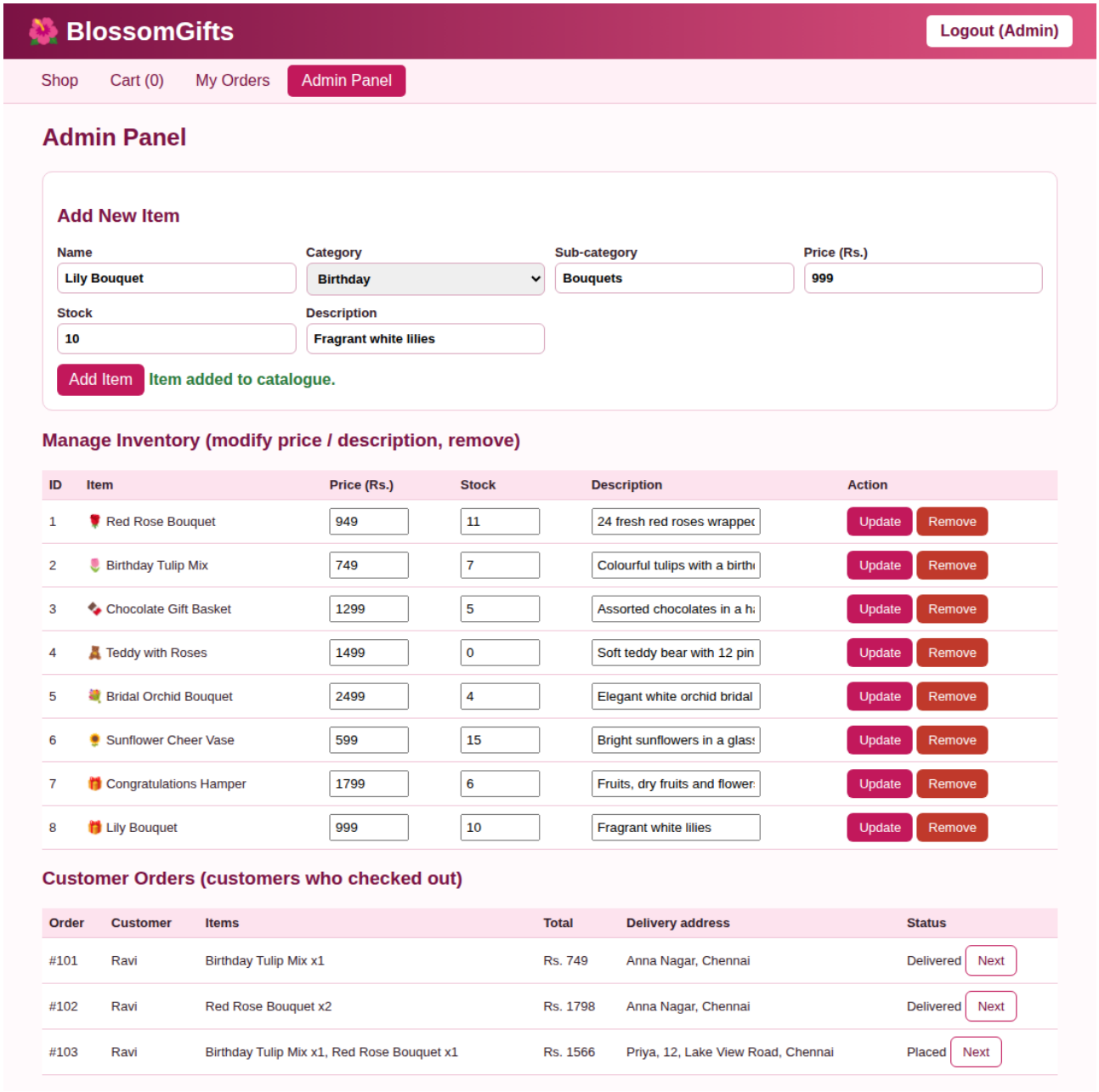
Shopping cart with quantity edit/remove, loyalty discount and delivery date/time details



Order placed - confirmation message after checkout



My Orders page with order tracking status



Admin panel: add item, modify price/description, remove items and view customers who checked out

RESULT

The BlossomGifts online flower and gift ordering website was developed using HTML, CSS and JavaScript. Customers could browse categories, search and filter items, add them to the cart, schedule delivery and track orders, and the administrator could add or remove items, change prices and descriptions and view the details of customers who checked out. The output was verified successfully.